

Solution to Problem 1. We construct a Markov chain with state space $S = \{0, 1, 2, 3\}$. We let $X_n = 0$ if an arrival occurs at time n . Also, we let $X_n = i$ if the last arrival up to time n occurred at time $n - i$, for $i = 1, 2, 3$. Given that $X_n = 0$, there is probability 0.2 that the next arrival occurs at time $n + 1$, so that $p_{00} = 0.2$, and $p_{01} = 0.8$. Given that $X_n = 1$, the last arrival occurred at time $n - 1$, and there is zero probability of an arrival at time $n + 1$, so that $p_{12} = 1$. Given that $X_n = 2$, the last arrival occurred at time $n - 2$. We then have

$$\begin{aligned} p_{20} &= \mathbf{P}(X_{n+1} = 0 \mid X_n = 2) \\ &= \mathbf{P}(T = 3 \mid T \geq 3) \\ &= \frac{\mathbf{P}(T = 3)}{\mathbf{P}(T \geq 3)} \\ &= \frac{3}{8}, \end{aligned}$$

and $p_{23} = 5/8$. Finally, given that $X_n = 3$, an arrival is guaranteed at time $n + 1$, so that $p_{30} = 1$.

Solution to Problem 2. We use a Markov chain model with 3 states, H , M , and E , where the state reflects the difficulty of the most recent exam. We are given the transition probabilities

$$\begin{bmatrix} r_{HH} & r_{HM} & r_{HE} \\ r_{MH} & r_{MM} & r_{ME} \\ r_{EH} & r_{EM} & r_{EE} \end{bmatrix} = \begin{bmatrix} 0 & .5 & .5 \\ .25 & .5 & .25 \\ .25 & .25 & .5 \end{bmatrix}.$$

It is easy to see that our Markov chain has a single recurrent class, which is aperiodic. The balance equations take the form

$$\begin{aligned} \pi_1 &= \frac{1}{4}(\pi_2 + \pi_3), \\ \pi_2 &= \frac{1}{2}(\pi_1 + \pi_2) + \frac{1}{4}\pi_3, \\ \pi_3 &= \frac{1}{2}(\pi_1 + \pi_3) + \frac{1}{4}\pi_2, \end{aligned}$$

and solving these with the constraint $\sum_i \pi_i = 1$ gives

$$\pi_1 = \frac{1}{5}, \quad \pi_2 = \pi_3 = \frac{2}{5}.$$

Solution to Problem 3. (a) The local balance equations take the form

$$0.6\pi_1 = 0.3\pi_2, \quad 0.2\pi_2 = 0.2\pi_3.$$

They can be solved, together with the normalization equation, to yield

$$\pi_1 = \frac{1}{5}, \quad \pi_2 = \pi_3 = \frac{2}{5}.$$

(b) The probability that the first transition is a birth is

$$0.6\pi_1 + 0.2\pi_2 = \frac{0.6}{5} + \frac{0.2 \cdot 2}{5} = \frac{1}{5}.$$

(c) If the state is 1, which happens with probability $1/5$, the first change of state is certain to be a birth. If the state is 2, which happens with probability $2/5$, the probability that the first change of state is a birth is equal to $0.2/(0.3 + 0.2) = 2/5$. Finally, if the state is 3, the probability that the first change of state is a birth is equal to 0. Thus, the probability that the first change of state that we observe is a birth is equal to

$$1 \cdot \frac{1}{5} + \frac{2}{5} \cdot \frac{2}{5} = \frac{9}{25}.$$

(d) We have

$$\begin{aligned} \mathbf{P}(\text{state was 2} \mid \text{first transition is a birth}) &= \frac{\mathbf{P}(\text{state was 2 and first transition is a birth})}{\mathbf{P}(\text{first transition is a birth})} \\ &= \frac{\pi_2 \cdot 0.2}{1/5} = \frac{2}{5}. \end{aligned}$$

(e) As shown in part (c), the probability that the first change of state is a birth is $9/25$. Furthermore, the probability that the state is 2 and the first change of state is a birth is $2\pi_2/5 = 4/25$. Therefore, the desired probability is

$$\frac{4/25}{9/25} = \frac{4}{9}.$$

(f) In a birth-death process, there must be as many births as there are deaths, plus or minus 1. Thus, the steady-state probability of births must be equal to the steady-state probability of deaths. Hence, in steady-state, half of the state changes are expected to be births. Therefore, the conditional probability that the first observed transition is a birth, given that it resulted in a change of state, is equal to $1/2$. This answer can also be obtained algebraically:

$$\mathbf{P}(\text{birth} \mid \text{change of state}) = \frac{\mathbf{P}(\text{birth})}{\mathbf{P}(\text{change of state})} = \frac{1/5}{\frac{1}{5} \cdot 0.6 + \frac{2}{5} \cdot 0.5 + \frac{2}{5} \cdot 0.2} = \frac{1/5}{2/5} = \frac{1}{2}.$$

(g) We have

$$\mathbf{P}(\text{leads to state 2} \mid \text{change}) = \frac{\mathbf{P}(\text{change that leads to state 2})}{\mathbf{P}(\text{change})} = \frac{\pi_1 \cdot 0.6 + \pi_3 \cdot 0.2}{2/5} = \frac{1}{2}.$$

This is intuitive because for every change of state that leads into state 2, there must be a subsequent change of state that leads away from state 2.

Solution to Problem 4 . (a) For $j < i$, we have $p_{ij} = 0$. Since the professor will continue to remember the highest ranking, even if he gets a lower ranking in a subsequent year, we have $p_{ii} = i/m$. Finally, for $j > i$, we have $p_{ij} = 1/m$, since the class is equally likely to receive any given rating.

(b) There is a positive probability that on any given year, the professor will receive the highest ranking, namely $1/m$. Therefore, state m is accessible from every other state. The only state accessible from state m is state m itself. Therefore, m is the only recurrent state, and all other states are transient.

(c) This question can be answered by finding the mean first passage time to the absorbing state m starting from i . It is simpler though to argue as follows: since the probability of achieving the highest ranking in a given year is $1/m$, independent of the current state, the required expected number of years is the expected number of trials to the first success in a Bernoulli process with success probability $1/m$. Thus, the expected number of years is m .

Solution to Problem 5 . (a) States 4 and 5 are transient, and all other states are recurrent. There are two recurrent classes. The class $\{1, 2, 3\}$ is aperiodic, and the class $\{6, 7\}$ is periodic.

(b) If the process starts at state 1, it stays within the aperiodic recurrent class $\{1, 2, 3\}$, and the n -step transition probabilities converge to steady-state probabilities π_i . We have $\pi_i = 0$ for $i \notin \{1, 2, 3\}$. The local balance equations take the form

$$\pi_1 = \pi_2, \quad \pi_2 = 6\pi_3.$$

Using also the normalization equation, we obtain

$$\pi_1 = \pi_2 = \frac{6}{13}, \quad \pi_3 = \frac{1}{13}.$$

(c) Because the class $\{6, 7\}$ is periodic, there are no steady-state probabilities. In particular, the sequence $r_{66}(n)$ alternates between 0 and 1, and does not converge.

(d) (i) The probability that the state increases by one during the first transition is equal to

$$0.5\pi_1 + 0.1\pi_2 = \frac{18}{65}.$$

(d) (ii) The probability that the process is in state 2 and that the state increases is

$$0.1\pi_2 = \frac{0.6}{13}.$$

Thus, the desired conditional probability is equal to

$$\frac{0.6/13}{18/65} = \frac{1}{6}.$$

(d) (iii) If the state is 1 (probability $6/13$), it is certain to increase at the first change of state. If the state is 2 (probability $6/13$), it has probability $1/6$ of increasing at the first change of state. Finally, if the state is 3, it cannot increase at the first change of state. Therefore, the probability that the state increases at the first change of state is equal to

$$\frac{6}{13} + \frac{1}{6} \cdot \frac{6}{13} = \frac{7}{13}.$$

(e) (i) Let a_4 and a_5 be the probability that the class $\{1, 2, 3\}$ is eventually reached, starting from state 4 and 5, respectively. We have

$$\begin{aligned} a_4 &= 0.2 + 0.4a_4 + 0.2a_5, \\ a_5 &= 0.7a_4, \end{aligned}$$

which yields

$$a_4 = 0.2 + 0.4a_4 + 0.14a_4,$$

and $a_4 = 10/23$. Also, the probability that the class $\{6, 7\}$ is reached, starting from state 4, is $1 - (10/23) = 13/23$.

(e) (ii) Let μ_4 and μ_5 be the expected times until a recurrent state is reached, starting from state 4 and 5, respectively. We have

$$\begin{aligned} \mu_4 &= 1 + 0.4\mu_4 + 0.2\mu_5, \\ \mu_5 &= 1 + 0.7\mu_4. \end{aligned}$$

Substituting the second equation into the first, and solving for μ_4 , we obtain

$$\mu_4 = \frac{60}{23}.$$